Post-Training Reorganizes Causal Use of J-Space

Evidence from the OLMo 32B Base, Think, and Instruct Lineage

Development-tier synthesis for the J-space campaign – not a frozen Phase 4 claim record

Central result. On the tested in-context Bank-S tasks, the OLMo base checkpoint shows no span-safe J-specific dependence. The effect appears by the first observed Think checkpoint, remains at OLMo-3.1 Think, and is absent at the OLMo-3.1 Instruct sibling. The pattern survives a frozen base-lens frame and exact common-support fact pairing. Measured J-space capacity, however, changes little across the three post-trained checkpoints. The evidence therefore points more strongly to *training-dependent recruitment or routing of an existing thin verbalizable channel* than to the creation of a larger J-space.

What this document does and does not claim

It asks how post-training changes J-space geometry, causal utilization, and workspace-like organization. It does not identify the causal training stage, prove that the Think objective created a new representation, or license the phrase "selective global workspace." Every OLMo lineage result is **phase4-development**: known banks, development cohorts, no untouched Phase 4 confirmatory outcome.

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Primary sources: Phase 4 registry and report; frozen Phase 2/3 imports
Branch snapshot: `interp_jspace_part2`, development state through 2026-08-01
Document version: August 2026, full lineage rewrite

The comparison to Qwen and Gemma is used as triangulation. The primary subject is the OLMo training lineage: what changes along the observed Think path, what returns toward the base regime at the Instruct sibling, and which measurements remain stable while causal behavior moves.

Contents

Executive synthesis

The phrase "J-space formation" hides several distinct scientific questions. Post-training might enlarge the verbalizable subspace, change its coordinates, change which content enters it, change which downstream computations consume it, or change only the model's behavioral strategy. The current OLMo evidence separates these possibilities better than a single workspace score can.

Key takeaway

The strongest supported reading is **causal recruitment without measured capacity growth**. Across the measured post-trained OLMo checkpoints, sparse occupancy remains 2 and centered excess variance stays roughly 0.4–1.3%. Yet Bank-S span-safe J-specific dependence is near zero at base, negative at both Think checkpoints, and near zero at the Instruct sibling. The same pattern survives a frozen base-model lens and a common-support fact cohort. Training is therefore associated with a change in how the model *uses* J-readable content, not merely how much J-readable variance the assay can find.

Paper-facing claim ladder

Rung	Licensed statement	What would upgrade or kill it
Phase 3 confirmatory	Qwen has a span-safe, J-specific heavy tail beyond an exact matched control; the effect replicates on held-out families.	Already confirmed and replicated; interpretation remains bounded by prose/selectivity results.
Phase 4 development	OLMo Bank-S causal dependence localizes to the observed Think path and reverses at the Instruct sibling under both own and frozen-base coordinates.	Untouched Phase 4 family split and a frozen primary; intermediate checkpoints or controlled continuation for training-stage attribution.
Mechanistic inference	Post-training appears to change utilization/routing of a conserved thin channel more than its measured capacity.	Base capacity, common-corpus lens geometry, Bank W derivation/redundancy, and patching into downstream receivers.
Not licensed	"Think training creates a global workspace," "Instruct has no internal channel," or "Gemma has no workspace."	These statements outrun the design or fail the transport-validity gate.

The most paper-relevant consequence is not a universal model ordering. It is a more structural point: **verbalizable causal organization is a training-moderated property**. Qwen, the OLMo Think branch, and the OLMo Instruct sibling do not merely occupy different points on one capacity axis. They exhibit different relationships among output protection, accessibility, bridge content, composition, and causal use.

1 Question, design, and estimands

1.1 The observed lineage is a natural experiment, not a randomized one

Checkpoint	Role in the inference	Pinned model / coordinate information
OLMo-3 32B base	shared pretrained anchor; defines the common coordinate frame	revision `c2b61dae...`; base lens SHA `92f32e38...`
OLMo-3 32B Think (3.0 Think)	first observed endpoint after unassayed post-training stages	own lens and seed-paired frozen base lens
OLMo-3.1 32B Think	later observed endpoint on the Think path	revision `832c3f54...`; own and common frames
OLMo-3.1 32B Instruct	sibling endpoint under a distinct post-training recipe, not a temporal successor to Think	revision `ac0587e4...`; own and common frames

Table 1: The observed OLMo 32B graph. The release structure localizes where an effect becomes visible, but it does not isolate SFT, preference optimization, data mixture, response format, or inference mode.

The graph should be read as

$$\text{Think path:}\quad \text{Base} \longrightarrow \underbrace{\text{unassayed post-training stages}}_{\text{recipe not separated}} \longrightarrow \text{3.0 Think} \longrightarrow \text{3.1 Think,}$$

$$\text{Instruct path:}\quad \text{Base} \longrightarrow \underbrace{\text{unassayed distinct post-training recipe}}_{\text{not the Think continuation}} \longrightarrow \text{3.1 Instruct.}$$

Evidence status

The proper design label is a **matched-lineage natural experiment**: shared scale and architecture, a shared base, pinned revisions, common tasks, and two coordinate frames. “Think versus Instruct” remains a bundle of training and formatting differences. The analysis can localize a change to an observed branch; it cannot identify the causal objective.

1.2 The intervention quantity

Each checkpoint uses prospective, prefix-disjoint G5 capability scoring on the 972-row Bank F + Bank S battery, then the same seven-condition teacher-forced grid: baseline, span-safe J, exact instantaneous rank/energy matched control, label-protected J, protected-energy control, mechanics-random control, and logit-space label protection. The primary lineage quantity is the **span-safe J-specific effect**.

Let L_0 be baseline answer log probability, L_J the span-safe J result, and L_C the exact profile-consuming matched control. Then

$$\Delta_J = L_J - L_0, \qquad \Delta_C = L_C - L_0, \qquad S_J = \Delta_J - \Delta_C.$$

A negative S_J means that removing the lens-selected content hurts more than removing an instantaneous random subspace with the same achieved rank and removed energy. It is a causal specificity measure under this intervention, not a measure of model capability, concept count, or global workspace size.

1.3 Four axes that training can move independently

1. **Capability**: can the unablated model produce an accepted answer, and with what answer margin?

2. **Capacity**: how much centered residual variance is captured beyond matched random dictionaries, and at what sparse occupancy?
3. **Coordinates**: does the effect require a checkpoint-specific lens, or survive a frozen common frame?
4. **Causal utilization**: does selected J content carry downstream load beyond its exact geometric dose?

These axes define three different meanings of "formation":

$$\begin{aligned}
\text{capacity formation:} &\quad C_{\text{post}} > C_{\text{base}}, \\
\text{coordinate formation:} &\quad \text{the effect exists only in } D_{\text{post}}, \\
\text{causal recruitment:} &\quad U_{\text{post}}(T) \neq U_{\text{base}}(T) \text{ even when } C \text{ and the frame change little.}
\end{aligned}$$

The current evidence is strongest for the third.

1.4 Audit floor

Every lineage grid passed the same acceptance battery: baseline replay drift from G5 below 8.7×10^{-7} nats, 100% matched-rank agreement, maximum matched-energy relative error below 2.6×10^{-4}, and exactly zero selected/protected overlap in the span-safe arm `[p4-lineage-grid-*-dev-v*]`. The frame analyses use bit-identical condition orders and seed namespaces and reproduce all 100,000-draw family bootstrap distributions from their registered rows.

Interpretation trap

The common-lens path earned several withdrawals before it became interpretable. One version mishandled rank-clamped protected-energy sites; another fixed the metadata but silently changed every stochastic control by changing the evidence-id seed namespace. The live pair reruns the common frame in the own grid's scientific namespace. Coordinate comparisons are meaningful only because the controls now pair exactly.

2 The OLMo trajectory result

2.1 Capability context: the cohort is not behaviorally uniform

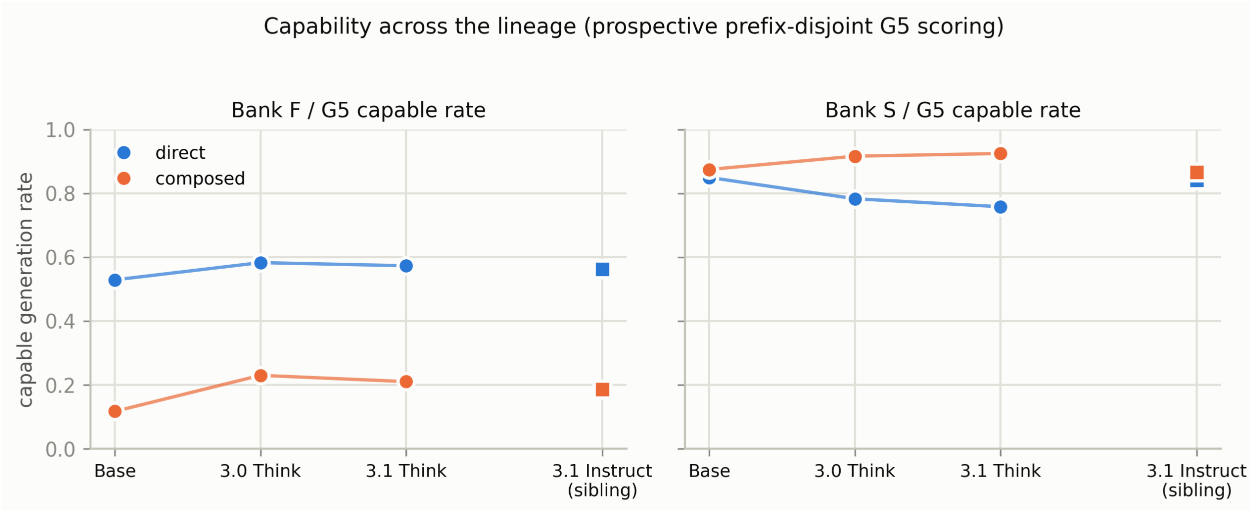

Figure 1: Prospective G5 capable-generation rates by checkpoint, bank, and variant. Squares mark the Instruct sibling. The Think path does not simply improve every task: Bank-S direct capability declines while Bank-S composed capability rises. Evidence: `p4-g5-bank-*-dev-v*`.

Three observations matter for interpreting intervention effects:

- Bank-F composition is hard at every checkpoint. The fact-bank development cohorts are direct-heavy, so Bank-F lineage claims remain imprecise and coordinate-sensitive.
- Bank-S direct capable rate falls from about 0.85 at base to 0.78 at 3.0 Think and 0.76 at 3.1 Think, while composed rises from about 0.88 to 0.92. The Instruct sibling sits nearer the base on both.
- Binary capability and conditional answer margin are not the same. On common-support facts, direct answer margin rises Base→3.0 Think by +1.95 nats and again by about +0.55 to 3.1 Think even though the binary direct-capable rate falls `[p4-lineage-common-cohort-analysis-olmo-dev-v1]`.

Capability membership was fixed before intervention outcomes at each checkpoint. The common-support closure in section 4 then asks whether the trajectory survives when the exact same facts are forced into the comparison.

2.2 Base: no span-safe content-specific dependence

At base, all four primary S_J cells are near zero: Bank-F direct +0.0245 [-0.0330, +0.0792], Bank-F composed +0.0145 [-0.1226, +0.1881], Bank-S direct +0.0005 [-0.0487, +0.0481], and Bank-S composed +0.0021 [-0.0309, +0.0385] `[p4-lineage-analysis-olmo3-base-dev-v1]`. The Bank-S composed-minus-direct contrast is +0.0016 [-0.0417, +0.0403].

This is not a dead assay. The same base grid produces a positive overall label-protected J-specific effect (+0.123) and nonzero mechanics-random and logit-protected effects with intervals excluding zero. The instrument can move the model; what is absent is dependence on the *span-safe lens-selected content* in these four cells.

2.3 Think path: direct dependence appears by 3.0 and persists at 3.1

By 3.0 Think, Bank-S direct S_J is -0.128 [-0.207, -0.049] in the own frame and -0.097 [-0.156, -0.045] in the frozen base frame. Bank-S composed is also negative in both frames. At 3.1 Think, direct dependence remains negative (-0.167 own; -0.155 common) while the composed-minus-direct contrast becomes precise and positive: $+0.118$ [+0.052, +0.188] own and $+0.117$ [+0.028, +0.209] common `[p4-lineage-analysis-olmo31-think-dev-v2,p4-lineage-trajectory-analysis-olmo-dev-v1]`.

The observed transition is not gradual in the available checkpoints. On paired common-support facts there is no resolved 3.0→3.1 Think increment. The main localization is the wide base→Think interval.

2.4 Instruct sibling: the Bank-S pattern returns toward zero

At OLMo-3.1 Instruct, Bank-S direct is -0.022 [-0.066, +0.018], composed is -0.018 [-0.076, +0.033], and their contrast is $+0.005$ [-0.041, +0.050] `[p4-lineage-analysis-olmo31-instruct-dev-v1]`. Every Bank-S interval includes zero in both the own and frozen-base frame. This is a sibling reversal, not evidence that Instruct has no internal content: Phase 3's frozen fact-item span audit found a substantial Instruct content residual on a different population. The defensible statement is task- and population-specific.

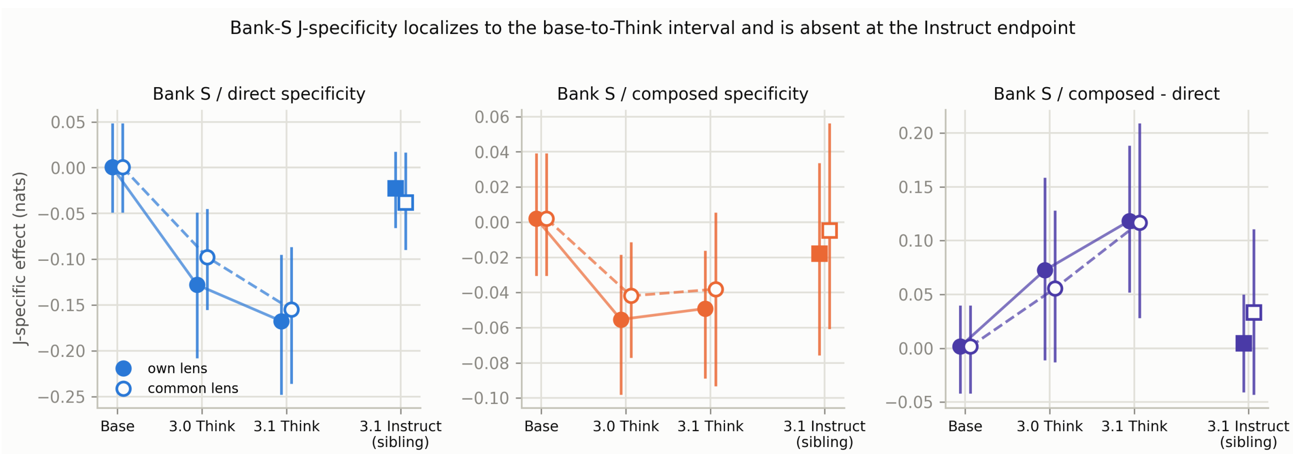

Figure 2: Bank-S J-specificity along the observed OLMo graph. Filled markers use each checkpoint's own lens; open markers use the frozen base lens. Lines connect only Base → 3.0 Think → 3.1 Think; the Instruct square is a sibling endpoint. Evidence: `p4-lineage-trajectory-analysis-olmo-dev-v1`.

2.5 A state-space view makes the training pattern clearer

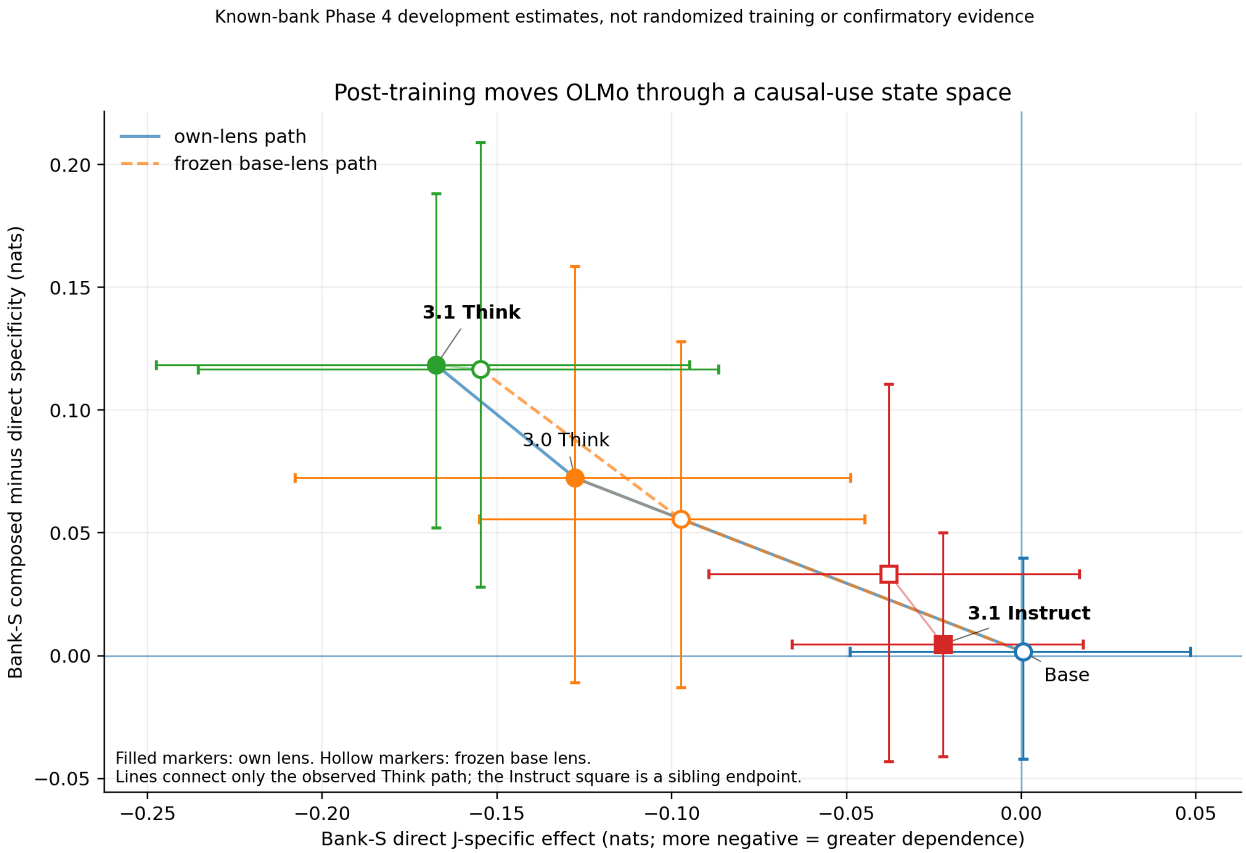

Figure 3: The lineage in a causal-use state space. Moving left means greater Bank-S direct dependence; moving up means composed prompts are relatively less dependent than direct prompts. The Think path leaves the base near the origin, then moves left and upward; the Instruct sibling returns near the origin. Small within-checkpoint connectors show own/common coordinate displacement. The figure is regenerated from the registered trajectory CSV by the supplied `olmo_lineage_synthesis_plots.py`.

Current synthesis

The simplest description of the observed trajectory is not "the workspace gets bigger." It is: **the Think branch recruits a thin J-readable channel for Bank-S direct state, while composed in-context state becomes relatively less dependent on that channel; the Instruct sibling does not show the same organization.** This is a statement about causal use under a specific assay and task population, not a statement that a consciousness-style workspace was created.

3 What post-training changes, and what stays nearly fixed

3.1 The moving arm is J-selected content, not matched dose

Because $S_J = \Delta_J - \Delta_C$, a trajectory could be manufactured by a wandering control. The raw arms rule that out.

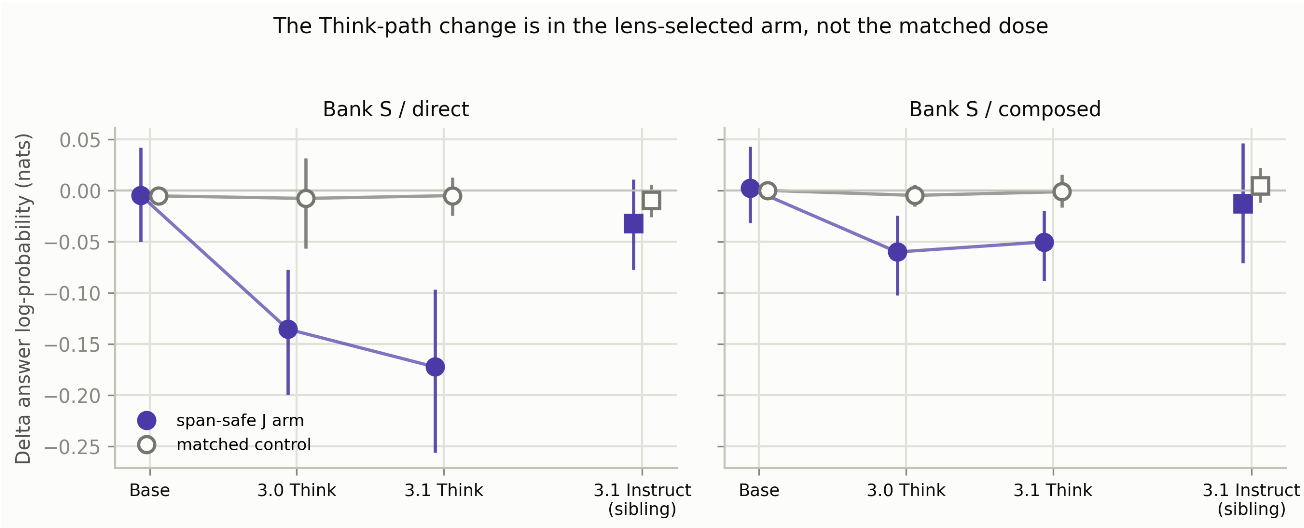

Figure 4: Raw Bank-S intervention effects before subtraction. The span-safe J arm moves from near zero at base to negative values at both Think checkpoints, then rebounds at Instruct. The exact matched control remains near zero in every cell. The training-path signal is carried by the selected content, not by a generic fragility to matched-rank, matched-energy lesions.

For direct Bank S, base→3.0 Think moves the J arm by roughly −0.13 nats while the control moves only a few thousandths. Think→Instruct reverses the J arm by about +0.14 with essentially no control movement. This rules out "Think is simply more lesion-sensitive" as the main account.

3.2 Measured capacity is almost unchanged across post-training

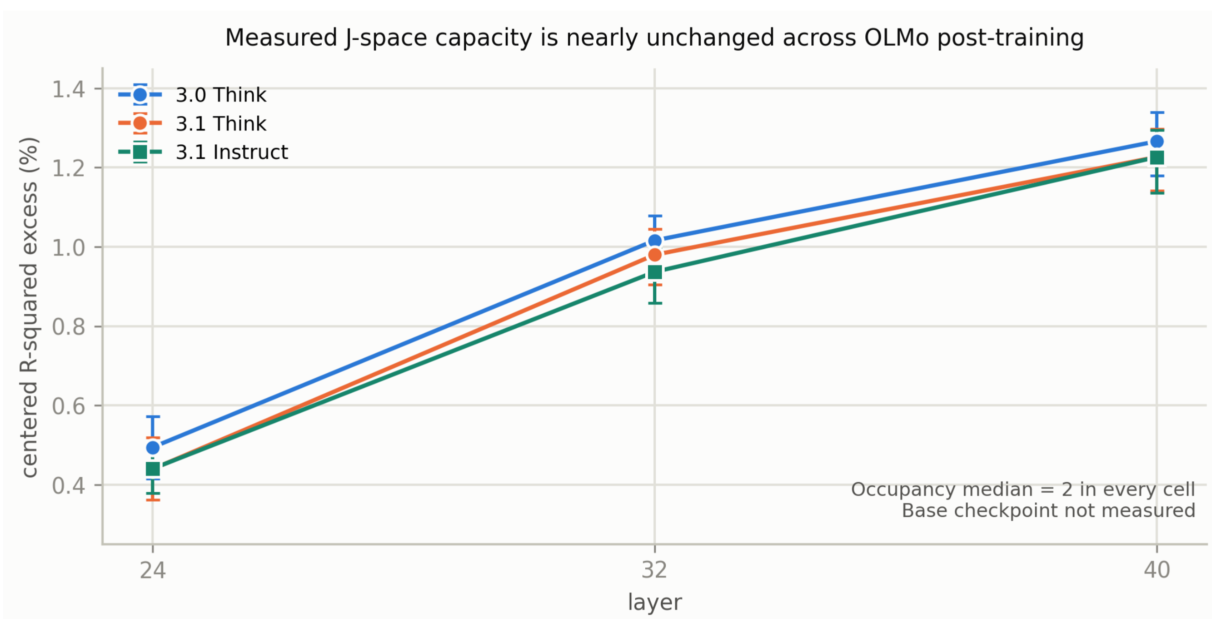

Figure 5: Paper-defined capacity for the three measured post-training OLMo checkpoints. Centered excess variance is approximately 0.44–0.49% at L24, 0.94–1.02% at L32, and 1.23–1.27% at L40; occupancy median is 2 in every cell. The base checkpoint has not yet been measured. Evidence: `r2-occupancy-*-v2`.

The 3.1 Think and Instruct capacity curves are nearly superposed even though their common-support Bank-S direct specificity differs by about 0.11–0.14 nats, depending on frame. The 3.0 Think checkpoint has the same occupancy and only slightly larger centered excess at these layers. Capacity and causal utilization are therefore empirically dissociated within this lineage.

Validity boundary

The base capacity cell is missing. "Capacity remains flat across post-training" is supported for 3.0 Think, 3.1 Think, and 3.1 Instruct. "Capacity did not change from base" is not yet measured and should not be written into the paper.

3.3 The effect does not require checkpoint-specific coordinates

The common-frame test applies the frozen base J-lens to every checkpoint while keeping stochastic controls paired. Bank-S effects preserve their sign and most of their interval structure. At 3.1 Think, the composed-minus-direct contrast is +0.118 in the own frame and +0.117 in the common frame; their paired difference is −0.002 [-0.055, +0.050]. At 3.0 Think, both direct and composed effects remain negative in both frames. At Instruct, both remain near zero.

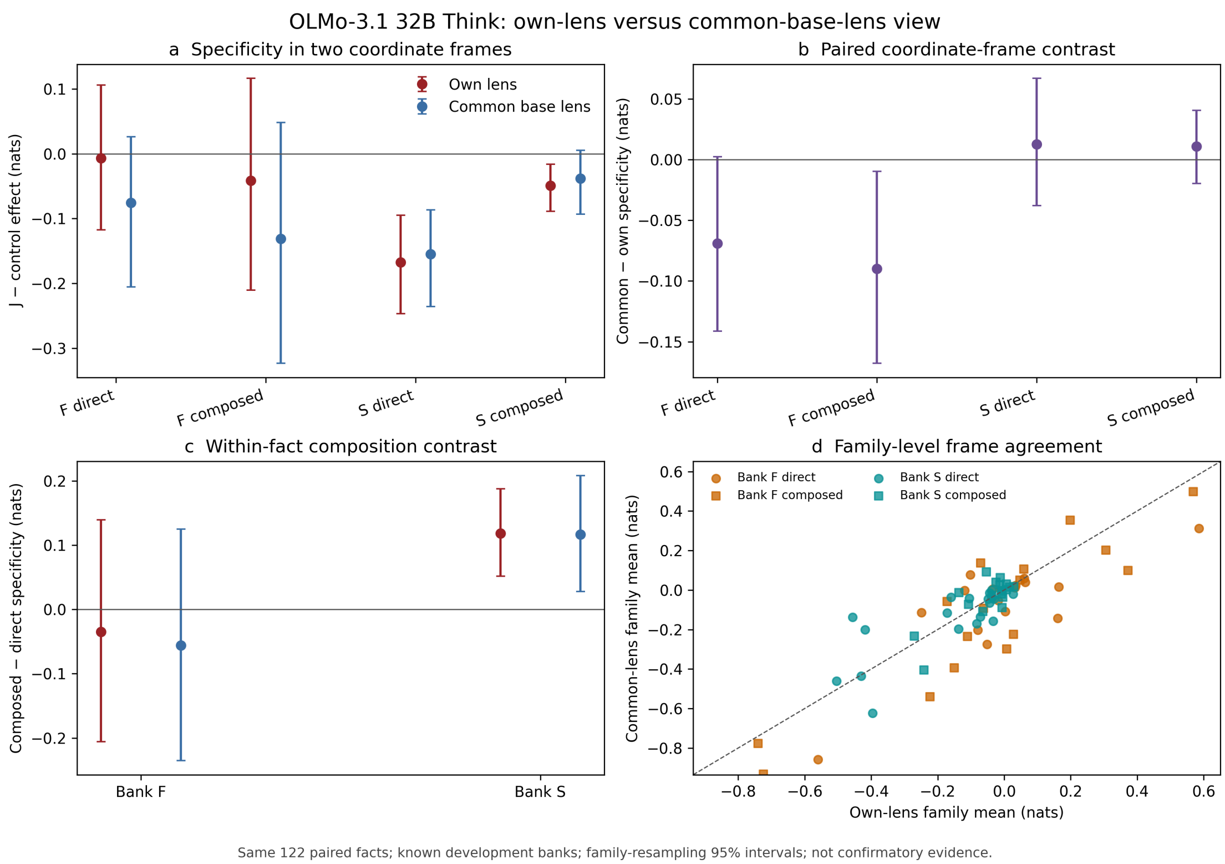

Figure 6: Seed-paired own-lens versus frozen-base-lens comparison at OLMo-3.1 Think `[p4-lens-frame-analysis-olmo31-think-dev-v1]`. Bank-S aggregates are stable across the two tested frames, while Bank-F contains the clearest coordinate sensitivity.

This does not mean the dictionaries are identical. Item-level own/common correlations are around 0.73–0.77 and mean absolute item shifts are 0.08–0.11 nats. The conclusion is narrower: checkpoint-specific lens refitting is not necessary to recover the aggregate Bank-S training pattern.

3.4 Readout coordinates transfer more readily than causal use

Part 2 applied the 3.0 Think lens unchanged to OLMo-3.1 Instruct. The donor and recipient retained the same 17/21 probe successes at rank ≤ 20, nearly identical pass@5/pass@20, median unembedding-row cosine 0.9984, and final-gain cosine 1.000. The transfer lost some rank-1 sharpening, but preserved the shortlist `[a0-transfer-gate-olmo31instruct-v1]`.

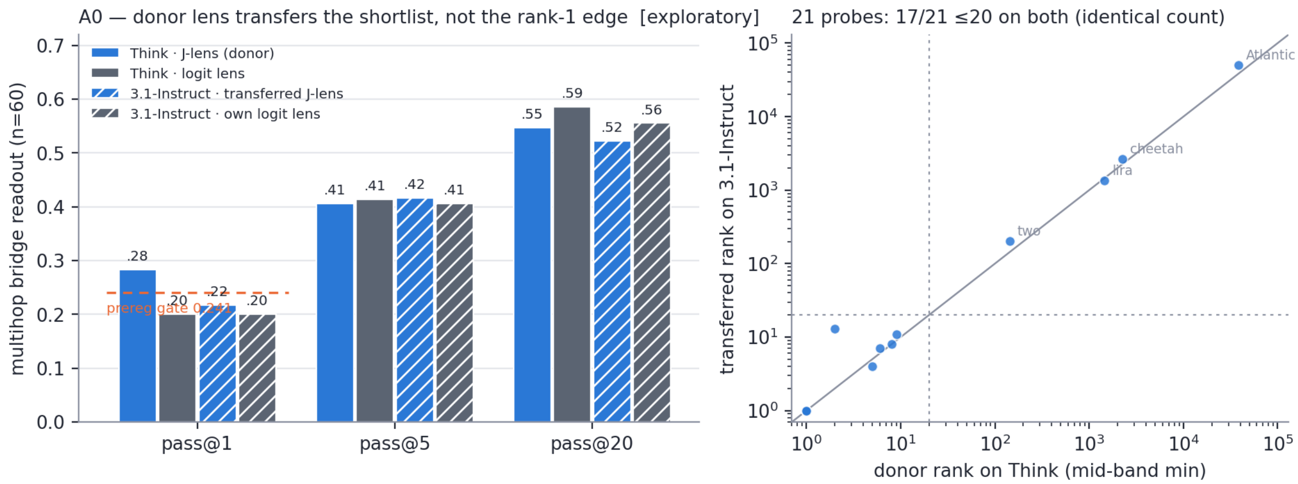

Figure 7: Exploratory donor-lens transfer. Post-training changes the sharp edge of the readout more than the broad shortlist. This is geometry/readout evidence, not a causal intervention comparison.

The conjunction is informative: the broad verbalizable coordinate system transfers, the measured capacity barely changes, and yet causal use on Bank S separates sharply. That is the empirical shape expected under **recruitment or rerouting of a conserved representation**.

4 Common-support closure: the same facts tell the same story

Checkpoint-specific capability cohorts could make a false trajectory if different facts survive at each model. The common-support analysis freezes membership from metadata only, before reading intervention outcomes. Seventy-nine facts across 25 families survive all four checkpoints; adjacent pairs retain 92–119 facts across 29–34 families `[p4-lineage-common-cohort-analysis-olmo-dev-v1]`.

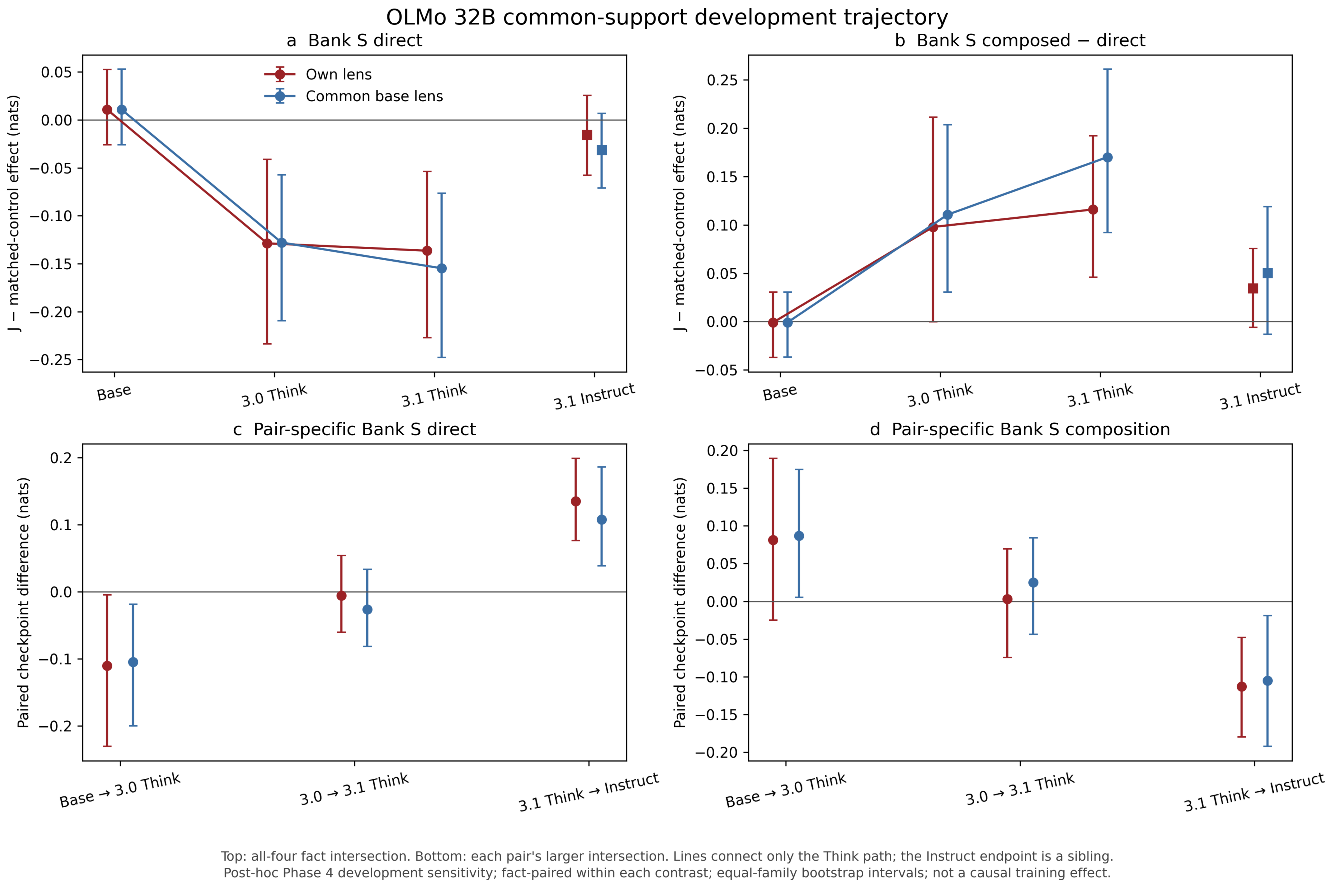

Figure 8: Post-hoc common-support sensitivity. The upper panels use the all-four intersection; the lower panels use larger adjacent-pair intersections. This removes changing fact composition as one explanation, but it does not randomize training.

Bank-S adjacent contrast	**Own direct**	**Common direct**	**Own composition**	**Common composition**
Base → 3.0 Think	−0.110*	−0.105*	+0.081	+0.087*
3.0 → 3.1 Think	−0.006	−0.027	+0.003	+0.025
3.1 Think → Instruct	+0.135*	+0.108*	−0.113*	−0.105*

Table 2: Right-minus-left checkpoint deltas on identical facts, equal-family weighting, 100,000 family bootstraps. * marks a 95% interval excluding zero. "Composition" is the change in composed-minus-direct specificity.

The direct base→Think decrease survives in both frames. The composition increase has the same sign in both frames and excludes zero in the common frame. No additional 3.0→3.1 Think change is resolved. The sibling Think→Instruct comparison reverses both direct dependence and the composition contrast in both frames, with all four intervals excluding zero. Every one-family-out estimate preserves the signs of the emergence and the reversal.

Baseline-answer-LP adjustment keeps the common-frame base→Think direct change below zero but widens several other intervals. Capability is therefore tracked as a moderator rather than declared irrelevant. The closure shows that changing fact membership is not sufficient to explain the trajectory; it does not prove the training recipe caused it.

5 What does "J-space formation" mean here?

A useful way to interpret the lineage is to separate three objects:

$$\underbrace{D_{\theta,\ell}}_{\text{verbalizable coordinates}}, \quad \underbrace{C_{\theta,\ell}}_{\text{capacity / occupancy}}, \quad \underbrace{U_{\theta,\ell,T}}_{\text{causal utilization on task } T}.$$

The fitted J dictionary D describes a coordinate frame. Capacity C asks how much activation variance can be represented sparsely beyond matched random controls. Utilization U asks whether the selected content is causally needed for a task after dose is matched.

Formation hypothesis	**Expected signature**	**Observed OLMo evidence**	**Current verdict**
Larger workspace forms	occupancy or centered excess rises on Think	Occupancy 2 and nearly overlapping post-training capacity curves; base unmeasured	not supported on measured post-training checkpoints
New coordinate system forms	effect appears only under own Think lens; donor/-common frame fails	Bank-S effect survives frozen base lens; broad donor shortlist transfers	not necessary for the aggregate effect
Existing channel is recruited	capacity and broad coordinates stable while S_J changes	J arm moves, matched control stays near zero; common-support and common-frame trajectory survive	best-supported development account
Behavioral strategy changes only	capability/output formatting changes but causal channel does not	capability changes, but exact matched intervention dependence also changes	insufficient as a complete account

Table 3: Different meanings of "workspace formation." The current evidence is most consistent with causal recruitment or rerouting, but the missing base capacity and intermediate checkpoints keep stronger wording out of reach.

This distinction matters for the main paper. A model can retain a small, readable coordinate system through post-training while changing which computations write into it, how long content persists, which downstream components read it, and whether prompt-supplied state substitutes for it. Those are organization changes even if concept count and explained variance stay flat.

6 Relation to the main paper, Qwen, and Gemma

6.1 The main paper's claim is a bundle, not one number

The original workspace interpretation combines several properties: limited capacity, verbalizable coordinates, broad downstream availability, causal importance for multi-step behavior, and relative sparing of ordinary language. This campaign supports some parts of that bundle more strongly than others. It has strong evidence for verbalizable causal channels and exact-control specificity. It does not currently support a selective global-workspace claim for OLMo: prose damage is not cleanly spared, and the lineage result is Bank-S/task-population specific.

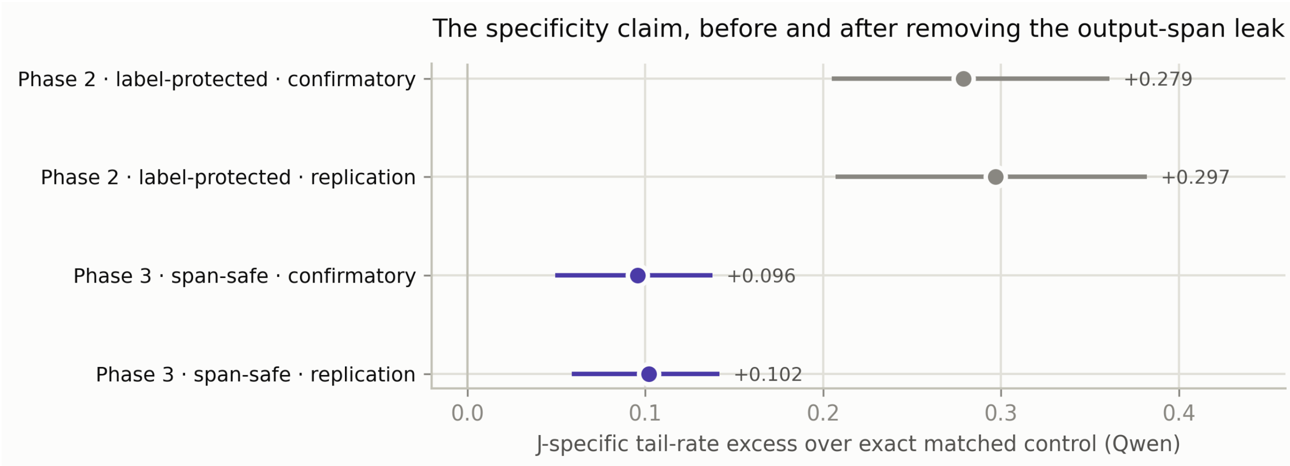

Figure 9: Qwen's specificity claim before and after output-span repair. The large label-protected Phase 2 tail shrinks under span-safe protection but survives and replicates in Phase 3. This is the confirmatory anchor against which the OLMo lineage development result is interpreted.

6.2 Qwen is a positive comparison, not merely a bigger OLMo

Qwen differs from OLMo on several independent axes:

- It has higher measured J-space capacity: roughly 5–6% centered excess and occupancy 3–4 under the published lens, versus OLMo's roughly 0.4–1.3% and occupancy 2.
- Its span-safe heavy tail rejects at confirmatory tier (+0.096) and replicates on held-out families (+0.102) `[p3-inference-audit-v1,p3-replication-analysis-v1]`.
- Label-protected and span-safe effects mostly rescale the same items ($r = 0.75$), while on OLMo they reallocate damage ($r = 0.20/0.29$).
- Qwen consumes a readable bridge route: true-bridge protection rescues, bridge-only lesion harms, and counterfactual substitution moves generation. OLMo Think does not show the same bridge mediation under the current assay.

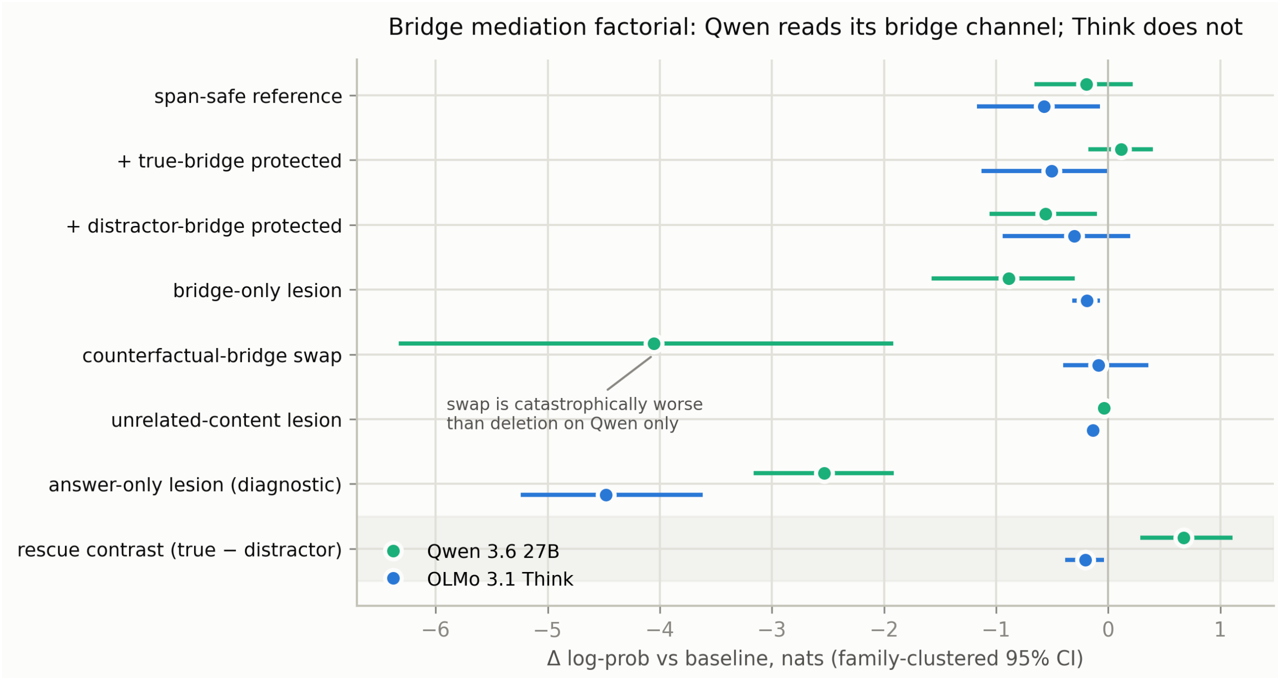

Figure 10: Development bridge mediation factorial. Qwen reads and follows the bridge channel; OLMo-3.1 Think does not show the same route. The dramatic Qwen swap is semantic, but Phase 4 still owes a bridge-versus-direct-answer-direction contrast on an adequately powered untouched bank.

Qwen therefore supplies a mechanistic contrast: a thicker, bridge-consumable knowledge-access channel rather than the accessibility-organized OLMo regime. It also supplies a methods warning. The Phase 4 same-corpus A120→A250 lens comparison is structurally strong in the L20–L44 assay band, but selected IDs and selected spans fail the frozen functional-stability thresholds. The precommitted branch continues to A500 `[p4-qwen-lens-convergence-drawA-n120-n250-dev-v1,p4-qwen-multilens-functional-gate-a120-a250-published-dev-v1]`. Readout similarity and causal selection stability are not interchangeable.

6.3 Gemma is not yet on the same scientific axis

The supplied Gemma-4 31B pilot is shaped differently for a more fundamental reason: the fixed linear transport premise appears to fail in the middle band. The reported response becomes less homogeneous and less additive as delivered perturbation scale grows, while the OLMo positive control approaches the linear doubling rule. Independent corpus slices also disagree strongly at the shallow edge. The appropriate current interpretation is not "Gemma lacks a workspace." It is "the mid-band fixed J-lens is not licensed on this checkpoint until an exact autograd-JVP versus finite-secant ladder closes the transport gate."

System	Transport / coordinates	Measured causal organization	Most defensible current noun
Claude paper	J-lens method used to define a limited verbalizable subspace; this campaign has not rerun the transport gate on Claude	reported multi-step causal dissociation with fluency sparing	global-workspace interpretation in the source paper
Qwen 3.6 27B	usable published lens; small-lens convergence still functionally unsettled	replicated span-safe specific tail; bridge-consumable route	knowledge-access channel with bridge organization
OLMo 32B lineage	partial but useful transport; broad frame transfer; thin capacity	Think-path Bank-S dependence, accessibility organization, no Qwen-like bridge consumption	training-moderated verbalizable causal channel
Gemma 4 31B pilot	fixed mid-band linear transport not yet licensed	causal J-space assay cannot be interpreted on equal footing	methods boundary / nonlinear-transport case study

Table 4: Cross-model comparison. Model differences are triangulation for the OLMo training question, not a substitute for a causal training intervention.

Gemma's candidate causes include context-dependent tangent cancellation, late basis conversion, attention routing transitions, repeated normalization curvature, and gated-MLP interactions. None is yet established. The exact JVP versus fp32 secant test, followed by layer and sublayer localization, is the scientifically prior experiment.

7 Mechanistic hypotheses for the OLMo post-training effect

These hypotheses are ordered by how directly they explain the lineage. Each is explicitly falsifiable.

H1 – Post-training recruits a conserved thin dictionary

The broad J-readable coordinates already exist at base and survive into both siblings. The Think recipe changes downstream consumption or persistence, not capacity. This matches the common-frame trajectory, donor-lens transfer, flat post-training capacity, and raw-arm decomposition.
Against it: aggregate frame robustness does not prove identical subspaces or receivers; item-level frame shifts remain nontrivial.
Decisive tests: fit every OLMo checkpoint on one frozen corpus, measure per-layer principal angles and top-k selection overlap, then patch Think activations or selected directions into Instruct under the common frame.

H2 – Think training makes prompt-supplied state substitute for internal state

The positive Bank-S composed-minus-direct contrast may mean that a composed prompt supplies an external scratchpad: the model relies less on its internal J channel when the state is repeated or explicitly derivable in the prompt. This would explain why direct Bank-S dependence is strongest at

Think while composed dependence is weaker.
Against it: the existing Bank-S templates confound composition, redundancy, and derivation.
Decisive test: Bank W's fully crossed load × supplied/derived × once/redundant factorial. The primary cell is internally derived, stated once; redundancy and supplied-state effects are named interactions rather than explanations added after the fact.

H3 – Instruct routes retrieval through output-adjacent geometry

The Instruct objective may optimize short direct answers, placing usable state closer to output preparation. That is compatible with accessibility organization, large label-arm damage on factual items, and near-zero Bank-S span-safe dependence. Instruct may contain internal content while offering less output-decoupled content for this assay to remove.
Against it: Phase 3 found the largest CI-clean span-safe content residual on Instruct factual items; the effect is population-dependent.
Decisive tests: vary the protected-output span, separate fact recall from in-context state, and localize receiver components under a fixed frame.

H4 – Accessibility, not workspace size, gates lesion cost

Across the OLMo pair, damage concentrates on less output-locked answers; Qwen reverses the sign. Think-path training may reorganize when retrieved content becomes committed to the answer rather than how many concepts the channel holds.
Against it: clean rank is already 1 for most lineage items, so the coarse rank statistic may miss relevant confidence and margin structure.
Decisive tests: continuous answer-margin and entropy models on the common-support cohort; matched-difficulty item construction; phase-resolved interventions before and after answer commitment.

H5 – One unobserved post-training stage installs the effect

The common-support data show an abrupt base→3.0 Think change and no resolved 3.0→3.1 increment. The change may arise at a particular SFT, preference, or reasoning stage rather than accumulate continuously.
Against it: released endpoints are too sparse to distinguish an abrupt transition from a smooth change hidden inside the long interval.
Decisive tests: intermediate checkpoints, or a controlled continuation from the base with the same evaluation frozen before training.

8 General geometry lessons

8.1 Capacity is not causal privilege

Within the OLMo post-trained trio, almost identical capacity coexists with substantially different Bank-S dependence. Across models, Qwen's higher capacity accompanies a cleaner bridge route, but three models are not a moderator analysis. Capacity is best treated as one structural coordinate, not as the explanation of causal organization.

8.2 A stable readout can support changing downstream use

A coordinate system can be conserved while the model changes which activations enter it, which components read it, and whether alternative routes compensate after ablation. This is analogous to retaining a shared interface while replacing the implementation behind it. The OLMo evidence most directly locates change downstream of broad coordinate availability.

8.3 Output protection changes the population of damaged items

On OLMo, label protection and true span protection damage different items. Removing output-span leakage reallocates rather than merely shrinks the lesion. That means post-training comparisons must name the protection geometry. A single "J-ablation effect" can mix output-adjacent dose and internal-content removal in model-dependent proportions.

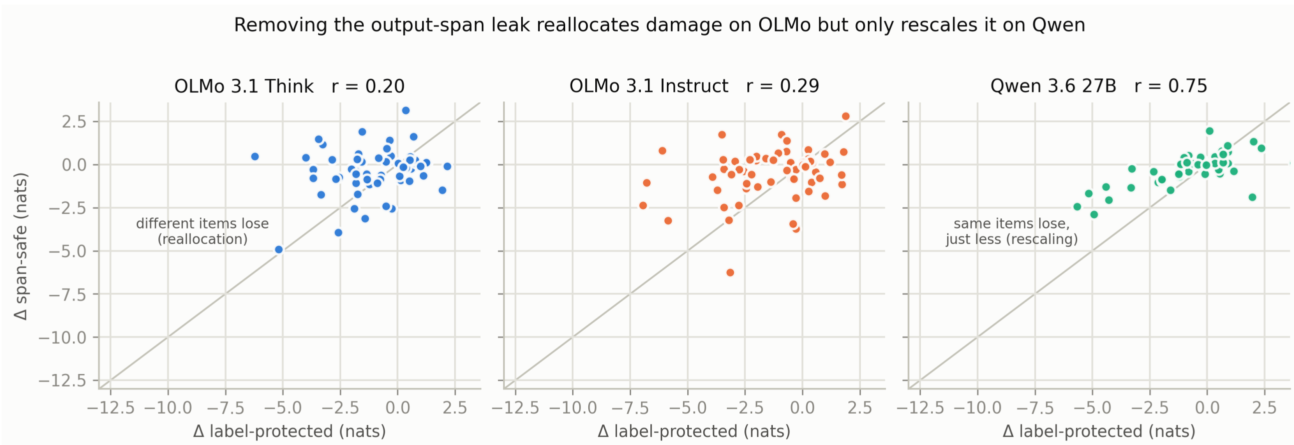

Figure 11: Label-protected versus span-safe effects. On OLMo the low correlations indicate reallocation; on Qwen the high correlation indicates rescaling. Output protection is part of the scientific object, not a minor implementation detail.

8.4 Transport validity and workspace interpretation are separate gates

OLMo can support a useful, if imperfect, fixed J coordinate through the assay band. Gemma may not. A negative workspace result is uninterpretable when the transport premise fails. Conversely, a readable lens is not by itself a causal license. Every model family needs a separate finite-dose and cross-context transport gate before it enters a shared J-space matrix.

9 Open questions and the next experiments

9.1 Highest-value OLMo experiments

1. **Run the Bank W capability and intervention gates on the OLMo pair first.** The 72-family factorial is authored, audited, and powered with a joint max-T primary at a 20-family minimum; capability and joint support remain pending `[p4-bank-w-candidate-v2,p4-bank-w-power-dev-v1]`. This is the cleanest adjudication of load, derivation, and redundancy, and the most direct test of the Think external-state-substitution account.

2. **Measure base capacity with the identical R2 estimator.** Without it, the experiment distinguishes post-trained utilization but cannot say whether the base→Think interval also changes readable capacity.
3. **Build a same-corpus lens-geometry trajectory.** Fit base, 3.0 Think, 3.1 Think, and 3.1 Instruct on one nested corpus and report matrix/row cosines, principal angles, selected-ID Jaccard, protected-span overlap, and functional stability. Common-frame behavior is not a substitute for direct geometry.
4. **Acquire intermediate post-training checkpoints or run a controlled continuation.** The current graph localizes the effect to a wide interval but cannot identify the responsible stage.
5. **Localize downstream receivers.** On the strongest Bank-S families, patch clean Think activations into ablated Think and into Instruct, then search attention/MLP receiver components with wrong-layer, unrelated, and dose-matched controls.
6. **Resolve generation phase.** Teacher-forced log probability is the clean primary, but Think training is fundamentally about a generation policy. Add prefill-only, reasoning-only, and final-answer interventions with fixed parser and truncation rules.

9.2 Current Phase 4 blockers and why they matter

- **Qwen lens convergence:** A120→A250 structure is strong, but selected-ID Jaccard and projector overlap fail frozen functional thresholds. The branch continues to A500. This blocks declaring a small Qwen lens canonical; it does not erase the published-lens Phase 3 results.
- **Qwen mode gate:** the first official thinking-on/off baseline failed because thinking-on truncated or failed to close on 7/20 families, leaving insufficient common support. A prospectively amended 2048-token gate is queued; no mode intervention claim exists yet.
- **Bank B bridge primary:** the verified 40-family candidate is dramatically underpowered for the proposed bridge-specific intersection- union endpoint. Reallocating the split cannot fix it. The design needs a new estimand or much larger bank, not a conveniently enlarged SESOI.
- **Phase 4 freeze:** the candidate preregistration remains unfrozen. No untouched confirmatory or replication outcome is licensed before protocol completion, independent review, PI sign-off, freeze commit, and tag.

9.3 Gemma remains a later methods leg

The next Gemma block should be an autopsy, not a model-matrix cell: exact fp32 JVP versus secant over a faithful ϵ ladder, intercept-plus-slope error fit, layerwise and attention/MLP localization, frozen attention-route controls, and local normalization/gate linearization. Only a passed transport band should receive a J-space causal assay.

10 Paper-facing conclusions and prohibited shortcuts

Recommended development-tier conclusion paragraph

Across four released checkpoints of the OLMo 32B lineage, span-safe J-specific dependence on in-context Bank-S state is absent at the base model, appears by the first observed Think checkpoint, persists at OLMo-3.1 Think, and is absent at the OLMo-3.1 Instruct sibling. The pattern survives a frozen base-model lens and exact common-support fact pairing, while measured J-space capacity is nearly unchanged across the three post-trained checkpoints. These results support a training-moderated causal-utilization account: post-training changes how a thin verbalizable channel is recruited or routed, rather than simply enlarging the measured channel. Because the checkpoints do not isolate the post-training stage and all results use known development banks, the evidence does not identify the Think objective as causal or establish workspace formation.

Allowed now	Not allowed now
The OLMo Bank-S dependence localizes to the observed Think path under the development assay.	Think training creates a global workspace.
The result survives two coordinate frames and common-support facts.	The J dictionaries are identical across checkpoints.
Measured post-training capacity is nearly flat while causal utilization moves.	Capacity did not change from base.
Instruct does not show the Think Bank-S organization on this population.	Instruct has no internal verbalizable channel.
Qwen and Gemma triangulate model-dependent organization and method validity.	Architecture alone explains Qwen, OLMo, or Gemma.

Table 5: Claim discipline for the lineage section of the eventual paper.

11 Data, evidence, and reproduction

11.1 Evidence map

Purpose	Registered evidence
Capability gates	`p4-g5-bank-*-dev-v*` (four OLMo checkpoints)
Own-lens grids and analyses	`p4-lineage-grid-*-dev-v1`; `p4-lineage-analysis-*`
Frozen base-lens grids	`p4-lineage-grid-*-common-base-lens-dev-v*`
Paired frame analyses	`p4-lens-frame-analysis-*`
Trajectory synthesis	`p4-lineage-trajectory-analysis-olmo-dev-v1`
Common-support closure	`p4-lineage-common-cohort-analysis-olmo-dev-v1`
Capacity and transfer	`r2-occupancy-*-v2`; `a0-transfer-gate-olmo31instruct-v1`
Phase 3 cross-model context	`p3-inference-audit-v1`; `p3-span-audit-cross-model-v1`; `p3-bridge-mediation-*`
Current methods blockers	`p4-qwen-multilens-functional-gate-*`; `p4-bank-w-power-dev-v1`; `p4-bank-b-design-feasibility-dev-v1`; `p4-qwen-mode-gate-dev-v1`

11.2 Regenerating the new synthesis figure

The supplied script reads the registered trajectory CSV and replots its stored point estimates and confidence limits. It does not recompute any bootstrap. From a Phase 4 run root:

```
python olmo_lineage_synthesis_plots.py \
  --run-root /content/drive/MyDrive/interpret/special-lab-1/phase4_20260731 \
  --out-dir ./figures
```

The expected input is `metrics/olmo-lineage-trajectory/trajectory_analysis/ p4-lineage-trajectory-analysis-olmo-dev-v1/ trajectory_table_olmo-lineage-trajectory.csv`. A direct `-trajectory-csv` path is also accepted.

Existing `p4f*` figures remain products of their event-registered repo producers. The attached `olmoL1-L4` figures are paper-local synthesis plots built from registered trajectory, G5, lineage-analysis, and R2 outputs.

A Per-checkpoint panels

These panels retain the full local context for each endpoint while the main text carries only the synthesis figures.

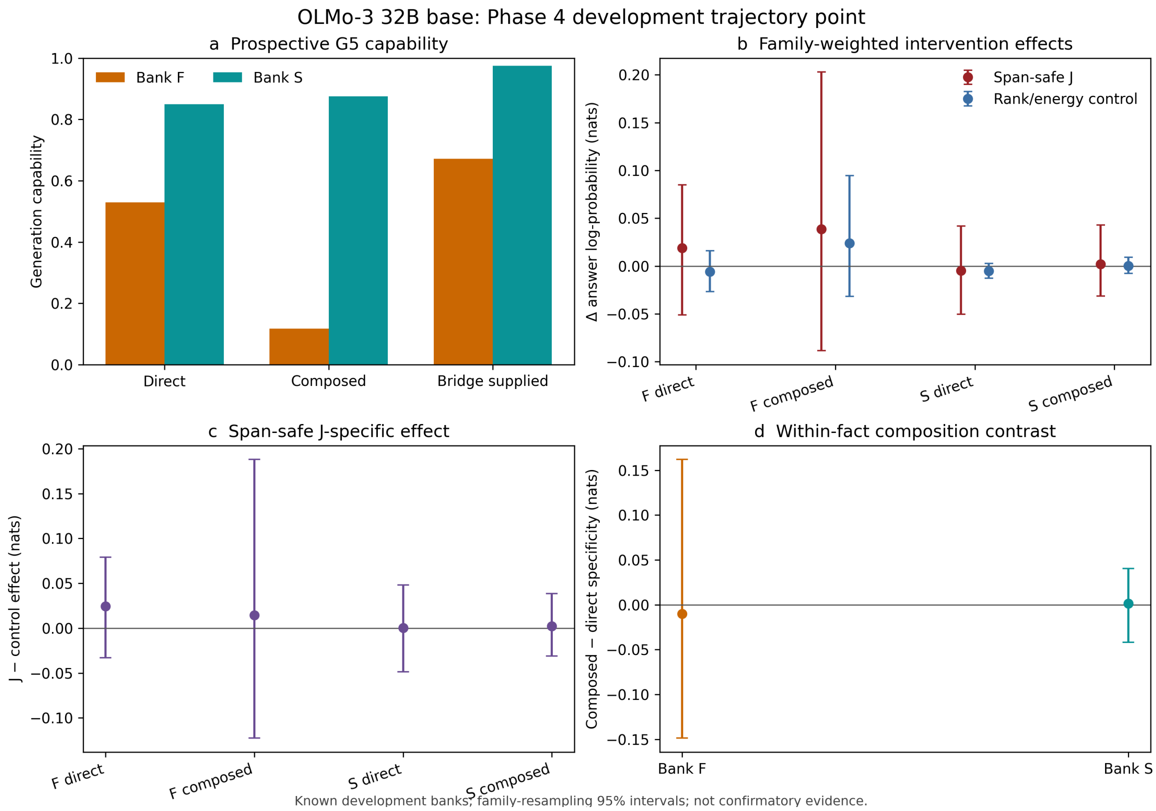

Figure 12: OLMo-3 32B base development point [p4-lineage-analysis-olmo3-base-dev-v1].

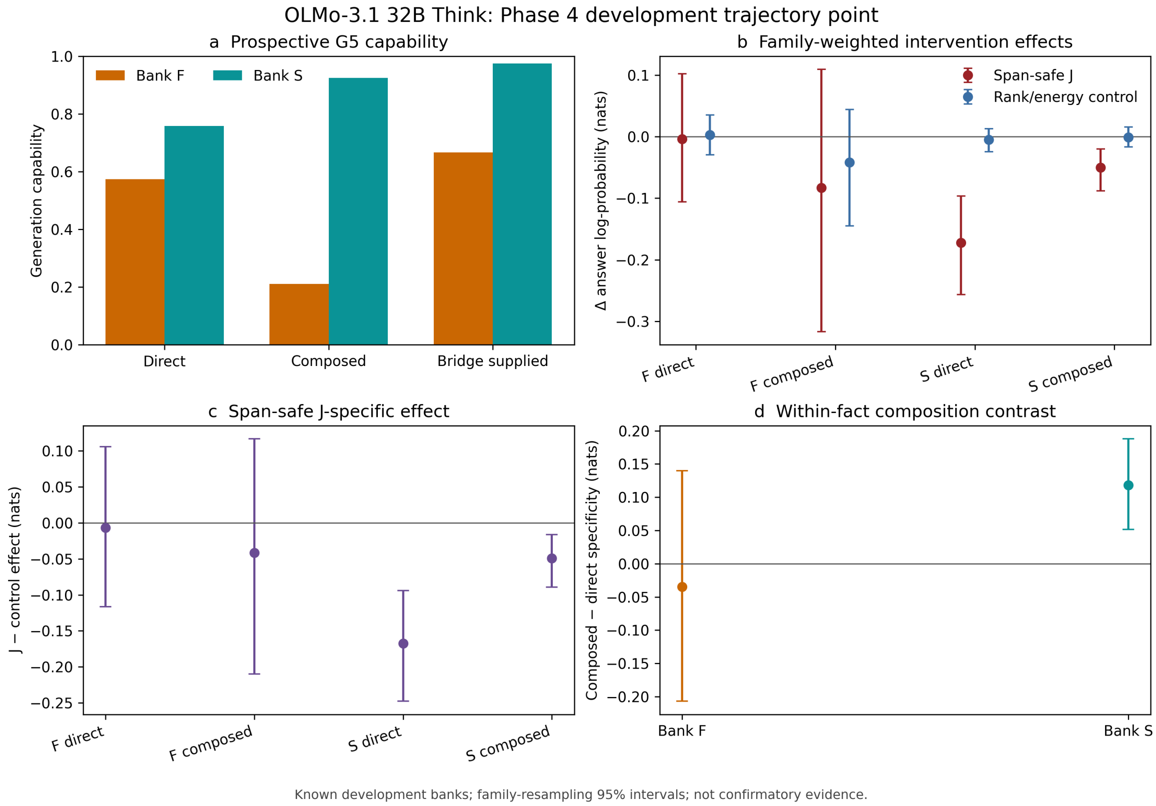

Figure 13: OLMo-3.1 32B Think own-lens development point [p4-lineage-analysis-olmo31-think-dev-v2].

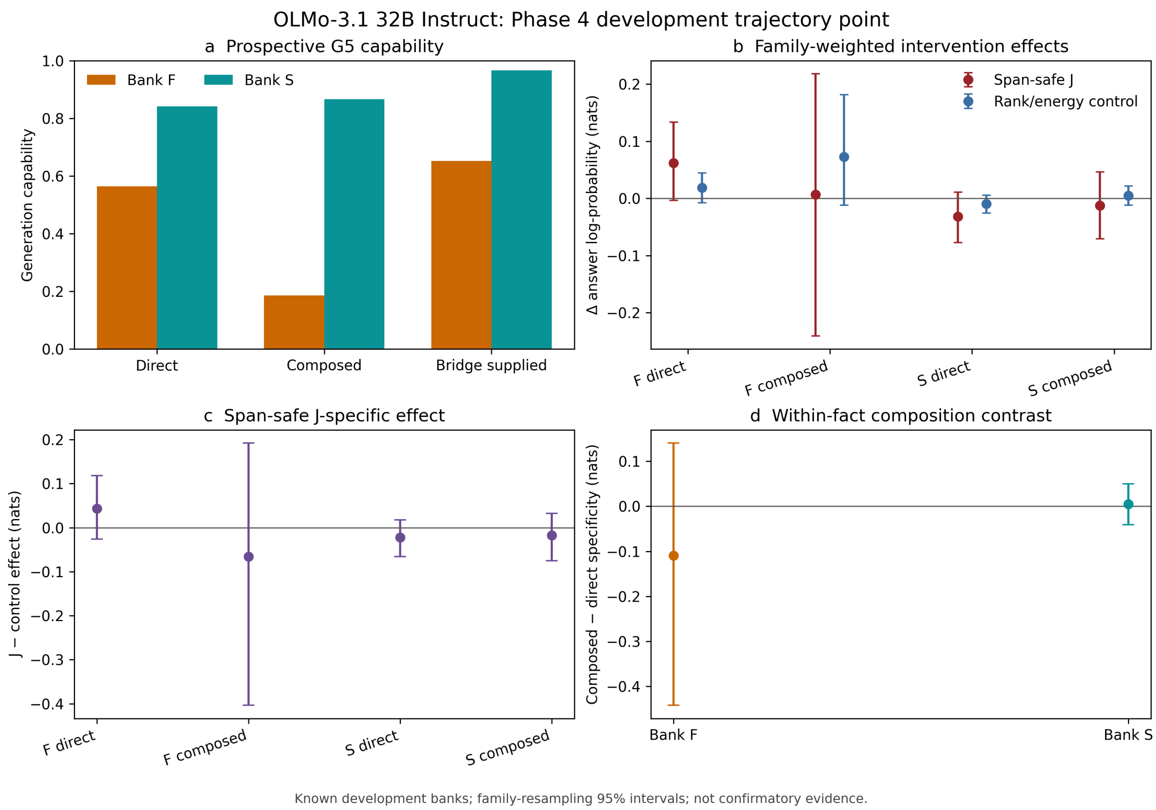

Figure 14: OLMo-3.1 32B Instruct own-lens development point [p4-lineage-analysis-olmo31-instruct-dev-v1].

B Earlier-phase context

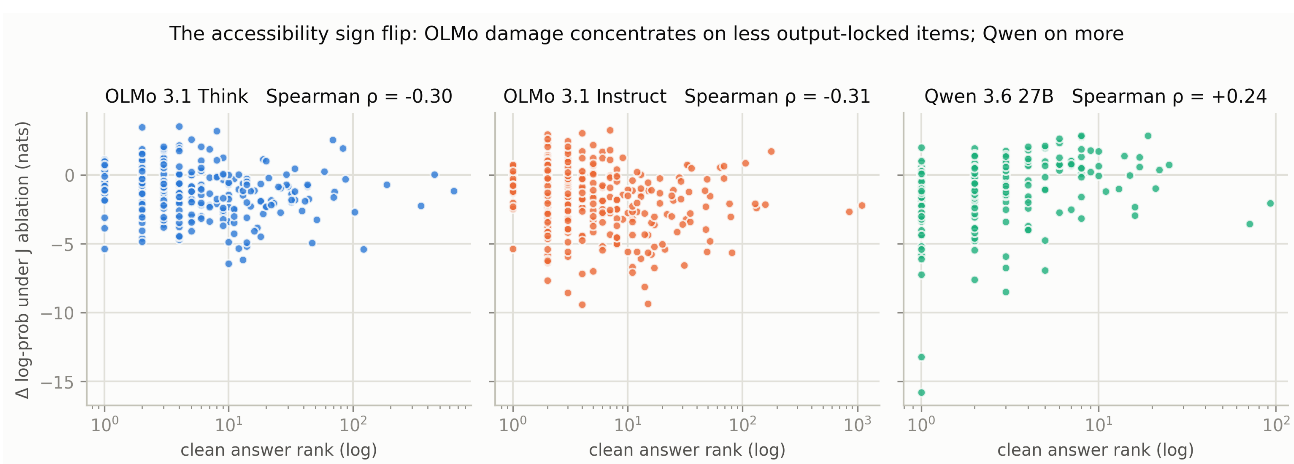

Figure 15: Phase 3 accessibility organization. OLMo damage concentrates on less output-locked items, while Qwen has the opposite sign [p3-overlap-mining-v1].

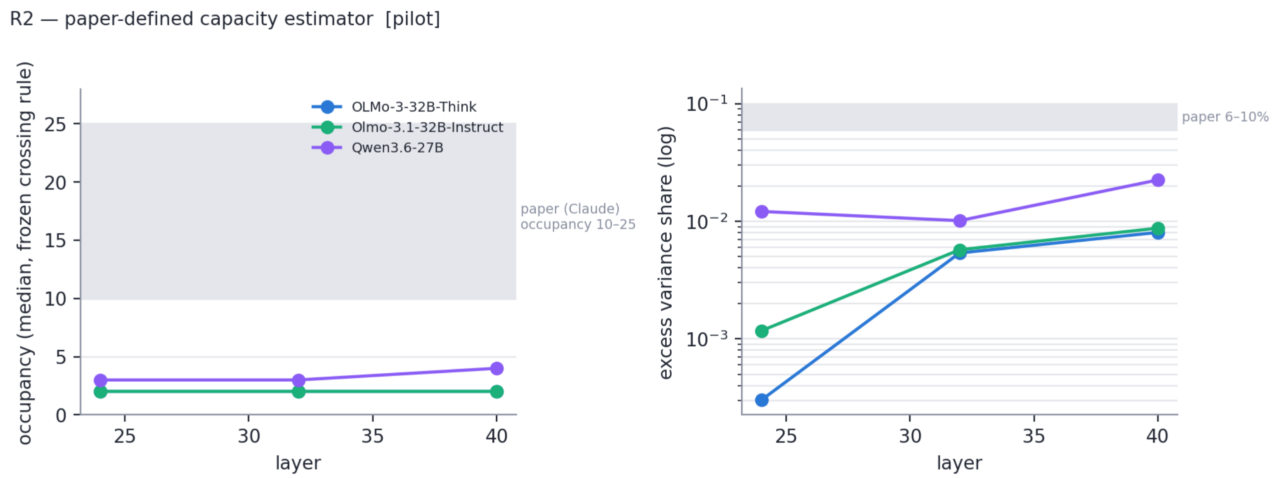

Figure 16: Earlier corrected capacity comparison. OLMo's occupancy is thin; Qwen sits nearer the lower edge of the source paper's excess-variance range. Later Phase 4 fit-size work adds a functional-convergence caveat to small-lens comparisons.

C Revision note

This rewrite removes the prior draft's meta-review from the main argument, places the checkpoint panels in an appendix, defines three distinct meanings of J-space formation, centers the capacity-versus-utilization dissociation, adds the common-support closure and current Phase 4 methods boundaries, and uses Qwen and Gemma as triangulation rather than as a flat model leaderboard. No registered OLMo point estimate was changed.